



BYD Microelectronics Co., Ltd.

BF20A6 Datasheet

1/10 inch VGA SOC CMOS Image Sensor With DVP and SPI

BF20A6 Datasheet



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1. General Description

The BF20A6 is a 1/10-inch VGA CMOS Image Sensor which includes CMOS image sensor (CIS) and ISP. It is fabricated with the world's most advanced CIS process to realize ultra-low dark noise, high sensitivity, high dynamic range and very low power imaging system. The sensor consists of a 640×480 pixel array which has an optical format of 1/10 inch for VGA. It has integrated noise canceling CDS circuits, analog global gain and separated R/G/B gain controller, automatic black level compensation and on-chip 10-bit ADC. The on-chip ISP provides a very smooth AE and accurate AWB control. It provides various data formats, such as Bayer RGB, YCbCr 4:2:2. It has a commonly used two-wire serial interface for host to control the operation of the whole sensor.

The product is capable of operating at up to 30 frames per second at 12MHz master clock in VGA mode, with complete user control over image quality and data format. All required image processing functions are also programmable through the two-wire serial bus.

2. Features

- Standard optical format of 1/10 inch for VGA
- 30 frames/sec VGA mode.
- Ultra-low dark noise at high temperature.
- Ultra-Low power consumption
TBD mW@30fps VGA
TBD uA @ power down
- Output formats: YUV422/Raw/ Only Y
- Power supply: 1.8/ 2.8V for I/O, 2.8V for VDD3A/VDD3D
- Horizontal /Vertical mirror.
- Auto black level control.
- Image processing function: Gamma Correction, Bad Pixel Correction, Color Interpolation, Low Pass Filter, Color Space Conversion, Color Correction, Edge Enhancement, Auto exposure, Auto White Balance, Color Saturation, Contrast, Skin Detection, and Data Format Conversion. Exposure control, Gain control, Test pattern, Image size control, Lens shading, Gamma and so on.
- Package: CSP.

3. Applications

- Cellular Phone Cameras
- Notebook PC cameras
- PDAs
- MP4
- Digital still cameras and camcorders
- Video telephony and conferencing equipment.

4. Technical Specifications

- Active pixel array: 640*480
- Pixel size: 2.25um×2.25um
- Sensitivity: TBD V/Lux-sec
- Dark current: TBD mV/s at 25°C
- Power supply: 2.8V/1.8V
- Power consumption: TBD mW@30fps (VGA)
- Standby current: TBD uA
- S/N Ratio: TBD dB
- Dynamic range: TBD dB
- Operating temperature: -10~65°C
- Stable Image temperature: 0~60°C
- Optimal lens chief ray angle: 28° non-linear
- Package: CSP

5. Functional Overview

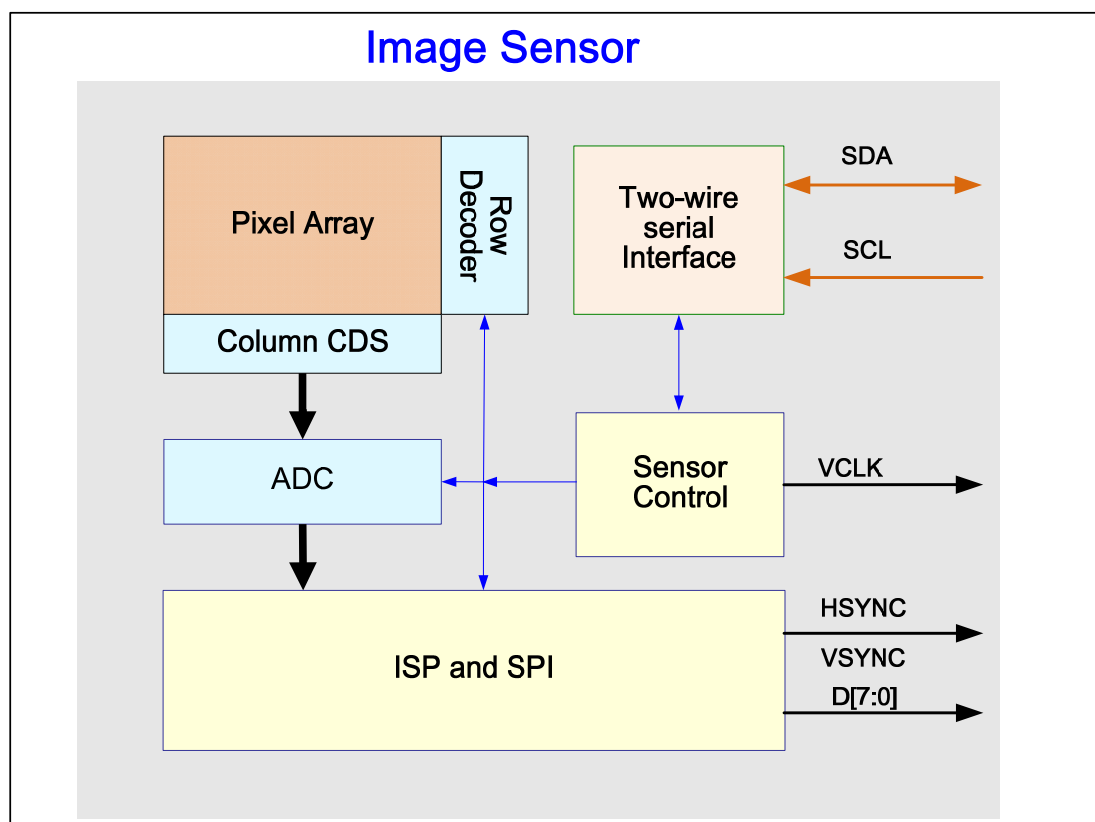


Figure 1. Block Diagram

BF20A6 has an active image array of 640x480 pixels. The active pixels are read out progressively through column/row driver circuits. In order to reduce fixed pattern noise, CDS circuits are adopted. The analog signal is

transferred to digital signal by A/D converter. It provide global gain to get accurate exposure under different light condition. The digital signals are controlled by Sensor control Block, including Exposure control, Gain control, Test pattern, Image size control and so on.

BF20A6 has on-chip PLL, it can be used by via two-wire serial interface bus setting.

5.1 Pixel Array

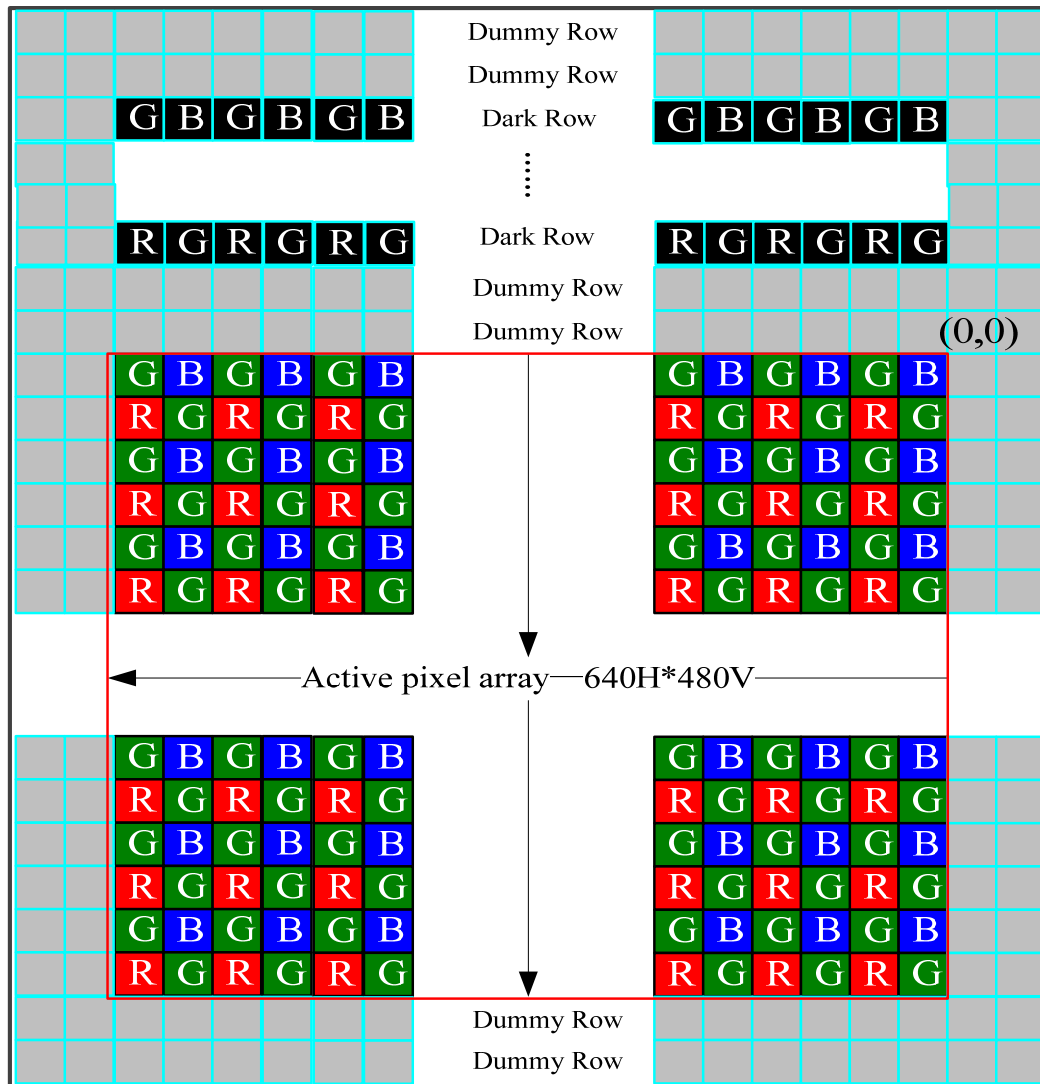


Figure 2. Sensor Array Region

The pixel array includes 640x480 effective pixels for imaging, whose address range is from (0, 0) to (639, 479). There are extra dark rows at the top side for black level control.

Pixel array is covered by Bayer color filters as can be seen in the figure2. The primary color BG/GR array is arranged in line-alternating fashion. Since each pixel can have only one type of color filter on it, only one color component can be obtained by a pixel. BF20A6 can provide the Raw Bayer data and YUV data (建议加上 Only Y) through SPI and DVP interface. If flip or not in column.

5.2 Column CDS

BF20A6 has column/row driver circuits to read out the pixel data progressively. The CDS (Correlated Double Sampling) circuit reduces temporal noise and pixel level FPN (Fixed Pattern Noise). The unique patented column buffer and ADC circuit remove column level FPN caused by various sources of manufacturing process variations.

5.3 Sensor control

- Array control and frame generation
- Internal timing signal generation and distribution
- Exposure control, Gain control, Test pattern,
- Image size control
- Frame rate timing

5.4 A/D converter

The analog signals are converted to digital forms and data are streamed out column by column. BF20A6 provides the 10-bit Raw Bayer data through an internal 10-bit data bus.

5.5 Automatic Black Control

The automatic black level control calculates the data of the dark row and controls the lowest black level for output image data.

6. Specifications

6.1 Electrical Characteristics

6.1.1 Absolute Maximum Ratings

- Supply voltage (VDDIO): 1.7 ~ 3.1 V
- Supply voltage (VDD3A/VDD3D): 2.7~ 3.1V
- Operating temperature: -10°C~65°C
- Storage temperature: -40°C~85°C
- ESD Rating, Human Body mode: 2000 V

Caution: Stresses exceeding the absolute maximum ratings may induce failure.

6.1.2 DC Parameters

Table 1. DC Operation Conditions

Symbol	Parameter	Unit	Min.	Typ.	Max.	Notes
VDDIO	I/O power supply	V	1.7	1.8	3.1	1
VDD3A	Analog power supply	V	2.7	2.8	3.1	—
VDD3D	Analog power supply	V	2.7	2.8	3.1	—
Vih	Input voltage logic “1”	V	0.7*VDDIO	—	—	—



Symbol	Parameter	Unit	Min.	Typ.	Max.	Notes
V _{il}	Input voltage logic "0"	V	–	–	0.2*VDDIO	–
V _{oh}	Output voltage logic "1"	V	0.9*VDDIO	–	–	–
V _{ol}	Output voltage logic "0"	V	–	–	0.1*VDDIO	–
I _{vddio}	VDDIO supply current, normal operation mode	mA	–	TBD	–	
I _{vdd3d}	VDD3D supply current,	mA	– –	TBD	–	2
I _{vdd3a}	VDD3A supply current,	mA	–	TBD	–	2

Note:

1. VDDIO=1.8V/2.8V (30 fps)
2. The Current of power is decided by the work mode, ex. Frequency of clock and output format. The Max. Current will not appear at the same time.

6.1.3 Clock Requirement

Table 2. AC Operation Conditions

Symbol	Parameter	Unit	Min.	Typ.	Max.	Notes
XCLK	External input clock	MHz	--	24	--	1
PCLK	Pixel clock of the system	MHz	--	12	--	2
VCLK	DVP Output clock	MHz	--	24	--	3
SCLK	two-wire serial interface clock frequency	KHz	--	400	--	4

Note:

1. XCLK is the input clock of system as the input of PLL.
2. PCLK is the pixel clock of system, and it is generated by PLL.
3. VCLK is the DVP output clock when the VCLK Pin is defined.
(Raw: VCLK=PCLK; YUV: VCLK=2*PCLK)
4. SCLK is driven by host processor. For the detail serial bus timing, refer to two-wire serial Interface section

6.2 Electro-Optical Characteristics

Clock frequency: 12MHz.

Operating voltage: VDDIO=2.8V, VDD3D=2.8V, VDD3A=2.8V.

Operating temperature: 25°C

Table 3. Electro-Optical Characteristics

Parameter	Unit	Min.	Typ.	Max.	Notes
Sensitivity	V/Lux·sec	--	TBD		1
Dark current	mV/sec	--	TBD	--	2
S/N ratio	dB	--	TBD	--	--
Dynamic Range	dB	--	TBD	--	--
Frame Rate	fps	--	30	30	3

Notes:

1. With color filter, measured at 50 lux green light condition at room temperature.
2. Measured at dark condition for exposure time of 1s (25 Celsius).
3. With 640×480 at PCLK 12MHz.

6.3 Timing

6.3.1 The inner VGA Fame Timing

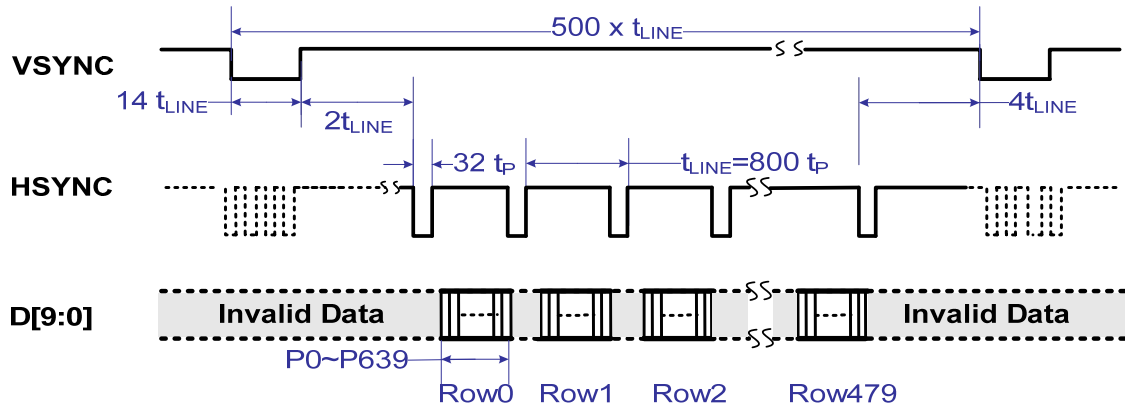


Figure 3. Inner VGA Frame Timing

Table 4. AC Characteristics

Symbol	Descriptions	Min.	Typ.	Max.	Unit
t_P	Pixel clock period	--	83.3	--	ns
f_{CLK}	Pixel Clock Frequency	--	12	--	MHz
t_{LINE}	Line length	--	$800 \times t_P$	--	ns

6.4 Color Filter Spectral Characteristics

The optical spectrum of color filters is shown below.

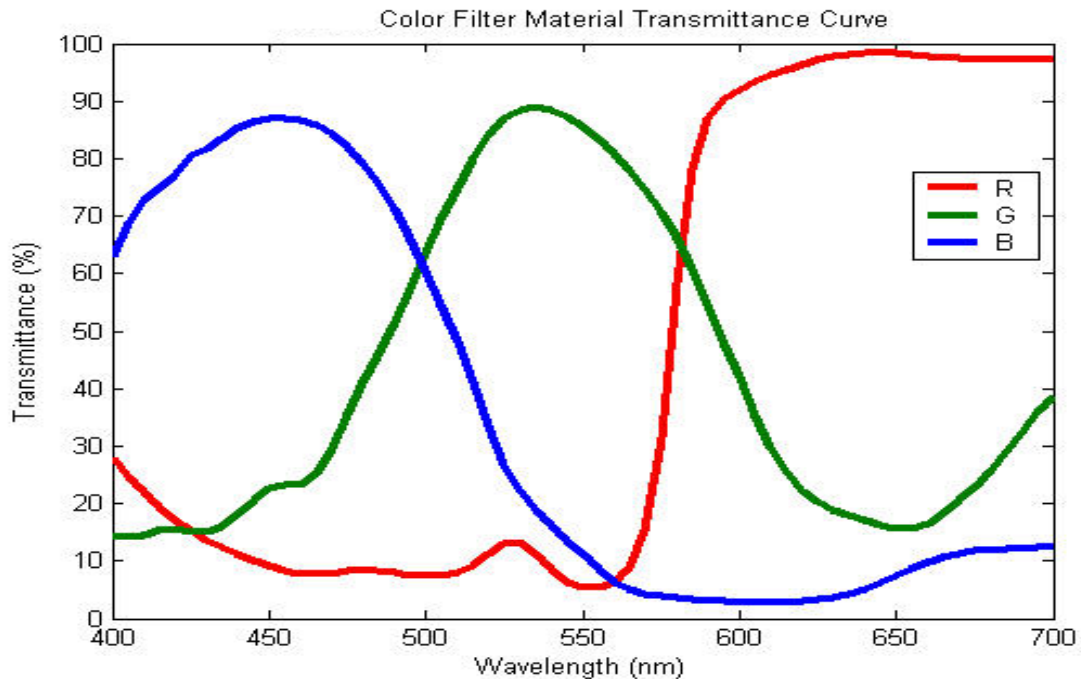


Figure 5. Spectral Characteristics

7. Two-wire serial interface& Register

7.1 Theory of Operation

The registers of BF20A6 are written and read through the two-wire serial interface. BF20A6 has two-wire serial interface slave. BF20A6 is controlled by the two-wire serial interface clock (SCLK), which is driven by the two-wire serial interface master. Data is transferred into and out of BF20A6 through the two-wire serial interface data (SDA) line. The SCL and SDA lines are pulled up to VDDIO by a 2kΩ off-chip resistor. Either the slave or the master device can pull the lines down. The two-wire serial interface protocol determines which device is allowed to pull the two lines down at any given time.

Note: Two-wire serial interface device address of BF20A6 is 7'b1101110 (0X6e), it doesn't include W/R bit.

Start bit

The start bit is defined as a HIGH to LOW transition of the data line while the clock line is HIGH.

Stop bit

The stop bit is defined as a LOW to HIGH transition of the data line while the clock line is HIGH.

Slave Address

The 8-bit address of a two-wire serial interface device consists of 7 bits of address and 1 bit of direction. A0 in the LSB of the address indicates write mode, and A1 indicates read-mode.

Data bit transfer

One data bit is transferred during each clock pulse. The two-wire serial interface clock pulse is provided by the master. The data must be stable during the HIGH period of the two-wire serial interface clock: it can only change when the two-wire serial interface clock is LOW. Data is transferred 8 bits at a time, followed by an acknowledge bit.

Acknowledge bit

The receiver generates the acknowledge clock pulse. The transmitter (which is the master when writing, or the slave when reading) releases the data line, and receiver indicates an acknowledge bit by pulling the data line low during the acknowledge clock pulse.

No-acknowledge bit

The no-acknowledge bit is generated when the data line is not pulled down by the receiver during the acknowledge clock pulse. A no-acknowledge bit is used to terminate a read sequence.

Sequence

A typical read or write sequence begins by the master sending a start bit. After start bit, the master sends the slave device's 8-bit address. The last bit of the address determines if the request will be a read or a write, where a 0 indicates a write and a 1 indicates a read. The slave device acknowledges its address by sending an acknowledge bit back to the master.

If the request was a write, the master then transfers the 8-bit register address to which a write should take place. The slave sends an acknowledge bit to indicate that the register address has been received. The master then transfers the data 8 bits at a time, with the slave sending an acknowledge bit after each 8 bits. The BF20A6 uses 8 bit data for its internal registers, thus requiring one 8-bit transfer to write to one register. After 8 bits are transferred, the register address is automatically incremented, so that the next 8 bits are written to the next register address. The master stops writing by sending a start or stop bit.

A typical read sequence is executed as follows. First the master sends the write-mode slave address and 8-bit register address just as in the write request. The master then sends a start bit and the read-mode slave address. The master then clocks out the register data 8 bits at a time. The master sends an acknowledge bit after each 8-bit transfer. The register address is auto-incremented after each 8 bit is

transferred. The data transfer is stopped when the master sends a no-acknowledge bit.

7.2 Two-wire Serial Interface Functional Description

Single Write Mode Operation



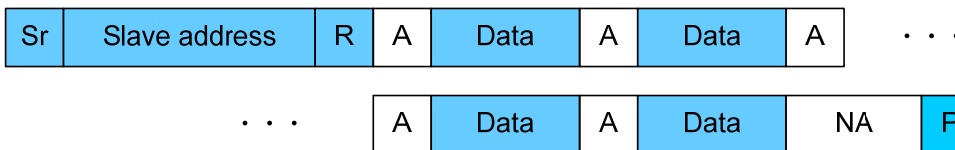
Multiple Write Mode (Register address is increased automatically)¹ operation



Single Read Mode Operation



Multiple Read Mode (Register address is increased automatically)¹ Operation



From master to slave

From slave to master

S: Start condition.

Sr: Repeated Start (Start without preceding stop.)

Slave Address:

write address = 0xdc = 11011100b

read address = 0xdd = 11011101b

R/W: Read/Write selection. High = read, LOW = write.

A: Acknowledge bit.

NA: No Acknowledge.

Data: 8-bit data

P: Stop condition

Note1: Continuous writing or reading without any interrupt increases the register address automatically. If the address is increased above valid register address range, further writing does not affect the chip operation in write mode. Data from invalid registers are undefined in read mode.

7.3 The Two-wire Serial Interface Timing

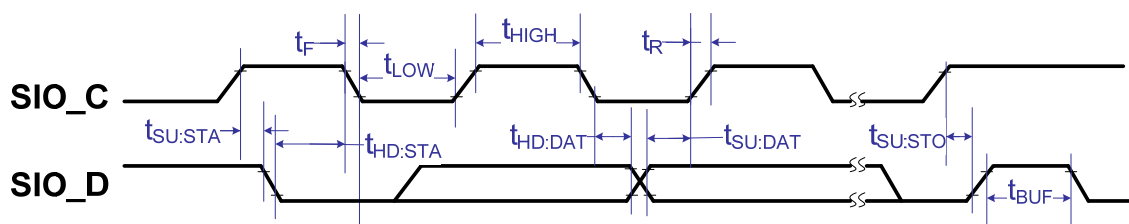


Figure 6. Two-Wire Serial Interface Timing

Table 5. AC Characteristics

Symbol	Descriptions	Min.	Typ.	Max.	Unit
t_R, t_F	two-wire serial interface rise/fall times	--	--	300	ns
t_{LOW}	Clock Low Period	1.3	--	--	us
t_{HIGH}	Clock High Period	600	--	--	ns
$t_{HD:STA}$	Start condition Hold Time	600	--	--	ns
$t_{SU:STA}$	Start condition Setup Time	600	--	--	ns
$t_{HD:DAT}$	Data-in Hold Time	0	--	--	ns
$t_{SU:DAT}$	Data-in Setup Time	100	--	--	ns
$t_{SU:STO}$	Stop condition Setup Time	600	--	--	ns

7.4 Register Summary (full list)

Table 6. BF20A6 all registers

Address	Name	Width	Default value	Description
00h	MODE_CNTL	8	23h	Bit[7]: BYPASS_BC 1'b1:Bypass digital black cntl. 1'b0:Don't bypass digital black cntl. Bit[6]: Reserved. Bit[5:4]: Dark row select. Bit[3]: Black level adjust mode. Bit[2]: Channel select. Bit[1]: Adjust mode. Bit[0]: Adjust mode.
01h	BLUE_GAIN	6	1ah	Blue gain 7bit for outdoor scene, 6 bits for indoor scene.
02h	RED_GAIN	6	1ah	Red gain 7bit for outdoor scene, 6 bits for indoor scene.
03h	VBLANK_PIX[12:8]	5	00h	Bit[7:5]: Reserved. Bit[4:0]:VBLANK_PIX[12:8].
04h	HSYST_LOCK	8	02h	Control the rising edged of HSYNC.
05h	HSYEN_LOCK	8	20h	Control the falling edged of HSYNC.
06h	DARKE_AVER	8	RO	Current black level
07h	DARKO_AVER	8	RO	Current black level
0bh	COM3	8	00h	Bit[7]: Adjust to CCIR packet code Bit[6]: Row counter after window Bit[5]: UV output value select. Bit[4]: DAT counter delay select. Bit[3:0]: Frame counter for skip.
0ch	AVER_LOCK	8	32h	Bit[7:4]: Lock value1 Bit[3:0]: Lock value2
0dh	H_HSYNC_EDGE_LOCK	8	00h	Bit[7:4]: HSYNC rising edge[11:8] Bit[3:0]: HSYNC falling edge[11:8]



Address	Name	Width	Default value	Description
0eh	<i>SH_Timing</i>	8	1fh	SH_Timing
0fh	<i>SH_Timing</i>	8	24h	SH_Timing
10h	<i>SH_Timing</i>	8	39h	SH_Timing
11h	<i>SH_Timing</i>	8	3ch	SH_Timing
12h	<i>COM0</i>	8	20h	Bit[7]:DAT_FOUR_SWAP. 1'b1:enable 1'b0:disable. Bit[6]:High byte and Low byte crossing-over of CODE: 1'b0: Low/High 1'b1: High/Low Bit[5]: Zhan xun or mtk ccir packet swap control : 1'b1:swap , 1'b0: no swap . Bit[4]: Zhan xun or mtk ccir packet select control: 1'b1:select zhan xun ccir ,1'b0:select mtk ccir Bit[3]: 1'b0: Output 8'h00 during hblank 1'b1: Output normally during hblank Bit[2]: Process RAW and bayer RAW Selection. 1'b0:bayer RAW 1'b1:process RAW Bit[1]: DOMAIN_SEL, for Only Y mode. Bit[0]: Raw RGB Selection. YCBYCR:x0. RAW:x1.
13h	<i>COM8</i>	8	05h	Bit[7] 1'b1:AE_TAR 1'b0:AE_TAR_MODIFY Bit[6]: Select INT_TIM. Bit[5]: AE adjusts rate. Bit[4]: The MSB of the minimum integral time for every step to avoid flicker for 50HZ/60HZ light. Bit[3]: GLB_GAIN0 written when AGC disable Bit[2]: AGC Enable. Bit[1]: Digital gain switch. 1'b0:OFF 1'b1:ON. Bit[0]: AEC Enable. 1'b0:OFF , 1'b1: ON.
14h	<i>P_PIXEL_OE</i>	8	RO	Reserved.
15h	<i>COM1</i>	8	00h	Bit[7]: VCLK reverse ,before gated latch Bit[6]: Hsync select before window 1'b0:no hsync when vsync_dat=0 1'b1:always has hsync Bit[5]: VSYNC output selection befor packet code, 1'b0:VSYNC_IMAGE. 1'b1:VSYNC_DAT Bit[4]: VCLK reverse, after gated latch Bit[3]: HSYNC output select befor code, 1'b0:HSYNC: 1'b1:HREF. Bit[2]: 1'b0:no HREF when VSYNC_DAT=0. 1'b1:always has HREF or not. Bit[1]: Negative image enable

Address	Name	Width	Default value	Description
				1'b0: Normal image. 1'b1: Negative image Bit[0]: Negative enable 1'b0: Normal 1'b1: Negative pixel
16h	COM2	8	70h	Bit[7]: Hsync mode after window: Bit[6]: Posedge and negedge of VSYNC in the MTK mode : 1'b1:enable. 1'b0:disable Bit[5]: Posedge of HSYNC in the MTK mode : 1'b1:enable 1'b0:disable Bit[4]: Posedge of HREF in the MTK mode : 1'b1:enable. 1'b0:disable Bit[3:2]: YUV Select. 2'b00:YUYV. 2'b01:UYVY. 2'b10:YVYU. 2'b11:VYUY. Bit[1]: CKGATE ahead Select. Bit[0]: GATED clock Select. 1'b0: Disable. 1'b1: Enable.
17h	HSTART	8	00h	Output Format-Horizontal Frame start high 8-bit
18h	HSTOP	8	a0h	Output Format-Horizontal Frame end high 8-bit
19h	VSTART	8	00h	Output Format-Vertical Frame start high 8-Bit
1ah	VSTOP	8	78h	Output Format-Vertical Frame end high 8-Bit
1bh	VHREF	8	00h	Bit[7:6]:Output Format-Vertical Frame(row)end low 2-Bit Bit[5:4]:Output Format-Vertical Frame(row)start low 2-Bit Bit[3:2]:Output Format-Horizontal Frame end low 2-bit Bit[1:0]:Output Format-Horizontal Frame start low 2-bit
1ch	MANU	8	80h	Manual U value
1dh	MANV	8	80h	Manual V value
1eh	SH_Timing	8	07h	SH_Timing
1fh	AVER_TAR_E	8	00h	Black level target for E col.
22h	AVER_TAR_O	8	00h	Black level target for E col.
23h	GLGAINREG	7	33h	Green Gain
24h	AE_TAR1	8	48h	Y target.
25h	K_GLB_GIAN AE_LOC	8	64h	Bit[7:4]:K_GLB_GIAN,LSB of GLB_GAIN slope. Bit[3:0]:AE_LOC,INT_TIME and GLB_GAIN lock.
26h	BLUE_GAIN_LOW_OUT	6	0dh	Bit[6:0]:The threshold of blue_gain_low_out
27h	BLUE_GAIN_HIGH_OUT	6	10h	Bit[6:0]:The threshold of blue_gain_high_out
28h	RED_GAIN_LOW_OUT	6	09h	Bit[6:0]:The threshold of red_gain_low_out
29h	RED_GAIN_HIGH_OUT	6	24h	Bit[6:0]:The threshold of red_gain_high_out



Address	Name	Width	Default value	Description
2ah	<i>SC_CNTL3</i>	8	98h	Bit[7]: 1'b1, pixel array reset interval also reset operate ; 1'b0, pixel array reset interval do nothing ; Bit[6]: Reset interval address select. Bit[5:0]: Reserved.
2bh	<i>VLANK_PIX[7:0]</i>	8	00h	Bit[7:0]:VLANK_PIX[7:0].
2ch	<i>AVER_E</i>	8	RO	Read out black aver for EB/EG
2dh	<i>AVER_O</i>	8	RO	Read out black aver for OR/OG
30h	<i>SH_Timing</i>	7	4dh	SH_Timing.
31h	<i>SH_Timing</i>	8	03h	SH_Timing.
32h	<i>SC_CNTL1</i>	8	00h	Bit[7]: 1'b1, VBLANK update netx frame. 1'b0, VBLANK update netx 2 frame. Bit[6:0]: SH_Timing.
33h	<i>SC_CNTL2</i>	8	26h	Bit[7:5]: Reserved. Bit[4]: 1'b1, GLB_GAIN_IN no delay. 1'b0, GLB_GAIN_IN delay one frame. Bit[3:0]: Reserved.
34h	<i>R_COEF</i>	8	46h	lens shading gain of R
35h	<i>G_COEF</i>	8	46h	lens shading gain of G
36h	<i>B_COEF</i>	8	46h	lens shading gain of B
39h	<i>PURE_TH</i>	8	58h	Bit[7:4]: Pure b threshold, multiplied by 4. Bit[3:0]: Pure r threshold, multiplied by 4.
3ah	<i>TSLB</i>	8	00h	Bit[7:2]:Reserved. Bit[1:0]:serial mode 2'b00:DVP 2'b01:1 bit 2'b10:2 bits 2'b11:4 bits.
3bh	<i>AWB_CB_LIM2</i>	8	18h	Bit[7]: Reserved. Bit[6:0]:AWB criterion: B, the threshold of B-G,for region 2.
3ch	<i>AWB_CR_LIM2</i>	8	18h	Bit[7]: Reserved. Bit[6:0]:AWB criterion: R, the threshold of R-G,for region 2.
3dh	<i>AWB_GR_LIM2</i>	8	18h	Bit[7]: Reserved. Bit[6:0]:AWB criterion: B, the threshold of G-B,for region 2.
3eh	<i>AWB_BR_LIM2</i>	8	18h	Bit[7]: Reserved. Bit[6:0]:AWB criterion: R, the threshold of G-R,for region 2.
3fh	<i>AWB_BR_LIM2</i>	8	28h	Bit[7]: Reserved. Bit[6:0]:AWB criterion: BR, the threshold of B+R-2*G ,for region 2.
40h	<i>k0</i>	8	24h	Gamma Correction Slop Coefficients
41h	<i>k1</i>	8	30h	Gamma Correction Slop Coefficients
42h	<i>k2</i>	8	24h	Gamma Correction Slop Coefficients
43h	<i>k3</i>	8	1dh	Gamma Correction Slop Coefficients
44h	<i>k4</i>	8	1ah	Gamma Correction Slop Coefficients
45h	<i>k5</i>	8	14h	Gamma Correction Slop Coefficients



Address	Name	Width	Default value	Description
46h	<i>k6</i>	8	11h	Gamma Correction Slop Coefficients
47h	<i>k7</i>	8	0eh	Gamma Correction Slop Coefficients
48h	<i>k8</i>	8	0dh	Gamma Correction Slop Coefficients
49h	<i>k9</i>	8	0ch	Gamma Correction Slop Coefficients
4ah	<i>SC_CNTL0D</i>	5	00h	Bit[7:5]: Reserved. Bit[4]: PDM_CNTL, 1'b0: POWERDOWN run free ; 1'b1: POWERDOWN Synchronous. Bit[3]: Mirror, 1'b0: Normal image, 1'b1: Mirror image. Bit[2]: Vertical Flip, 1'b0: Normal image, 1'b1: Vertically flip image. Bit[1:0]: Reserved.
4bh	<i>k10</i>	8	0bh	Gamma Correction Slop Coefficients
4ch	<i>k11</i>	8	09h	Gamma Correction Slop Coefficients
4eh	<i>k12</i>	8	09h	Gamma Correction Slop Coefficients
4fh	<i>k13</i>	8	08h	Gamma Correction Slop Coefficients
50h	<i>k14</i>	8	07h	Gamma Correction Slop Coefficients
51h	<i>TARGET1</i>	8	0dh	Bit[7:0]: Color Correction Coefficients low 8 bit (sign bits at TARGET_SIGN2[0]).
52h	<i>TARGET2</i>	8	0eh	Bit[7:0]: Color Correction Coefficients low 8 bit (sign bits at TARGET_SIGN2[1]).
53h	<i>TARGET3</i>	8	42h	Bit[7:0]: Color Correction Coefficients low 8 bit (sign bits at TARGET_SIGN2[2]).
54h	<i>TARGET4</i>	8	4ch	Bit[7:0]: Color Correction Coefficients low 8 bit (sign bits at TARGET_SIGN2[3]).
55h	<i>TARGET5</i>	8	76h	Bit[7:0]: Color Correction Coefficients low 8 bit (sign bits at TARGET_SIGN2[4]).
56h	<i>TARGET6</i>	8	21h	Bit[7:0]: Color Correction Coefficients low 8 bit (sign bits at TARGET_SIGN2[5]).
57h	<i>TARGET1_SK</i>	8	37h	Bit[7:0]: A light Color Correction Coefficients low 8 bit ({sign bits,hign 1 bit} at {TARGET_SIGN1[0],TARGET_H[0]}).
58h	<i>TARGET2_SK</i>	8	1eh	Bit[7:0]: A light Color Correction Coefficients low 8 bit ({sign bits,hign 1 bit} at {TARGET_SIGN1[1],TARGET_H[1]}).
59h	<i>TARGET3_SK</i>	8	afh	Bit[7:0]: A light Color Correction Coefficients low 8 bit ({sign bits,hign 1 bit} at {TARGET_SIGN1[2],TARGET_H[2]}).
5ah	<i>TARGET4_SK</i>	8	c5h	Bit[7:0]: A light Color Correction Coefficients low 8 bit ({sign bits,hign 1 bit} at {TARGET_SIGN1[3],TARGET_H[3]}).
5bh	<i>TARGET5_SK</i>	8	55h	Bit[7:0]: A light Color Correction Coefficients low 8 bit ({sign bits,hign 1 bit} at {TARGET_SIGN1[4],TARGET_H[4]}).
5ch	<i>TARGET6_SK</i>	8	18h	Bit[7:0]: A light Color Correction Coefficients low 8 bit ({sign bits,hign 1 bit} at {TARGET_SIGN1[5],TARGET_H[5]}).

Address	Name	Width	Default value	Description
5dh	TARGET_H	6	00h	Bit[5:0]: Control the highest value bit (bit[8]) of the A light coefficients. Bit[5]:TARGET6_SK. Bit[4]:TARGET5_SK. Bit[3]:TARGET4_SK. Bit[2]:TARGET3_SK. Bit[1]:TARGET2_SK. Bit[0]:TARGET1_SK.
5eh	GLB_GAIN_TH	8	20h	Bit[7]: Glb_gain adjust switch. 1'b1: OFF,do not be affect by glb_gain. 1'b0: ON,do adjust for glb_gain. Bit[5:0]: Used as glb_gain threshold for color martrix adjust, smaller value is for larger adjust.
5fh	AWB_SPEED OUTDOOR_EN GNGAINREG	6	21h	Bit[5:4]:AWB speed. Bit[3]: Bit[7]: outdoor_en ,outdoor model control. 1'b0:off 1'b1:on Bit[2:0]:G Channel Gain (Bit[2:0] is used as Green ain[5:3]).
60h	AVER_E_I2C	8	10h	Manual black level value for E col.
61h	AVER_O_I2C	8	10h	Manual black level value for O col.
62h	SH_Timing	8	7dh	SH_Timing.
63h	SH_Timing	7	0bh	SH_Timing.
64h	SH_Timing	7	1ah	SH_Timing.
65h	SH_Timing	7	34h	SH_Timing.
66h	SH_Timing	7	40h	SH_Timing.
67h	SH_Timing	8	9ah	SH_Timing.
68h	SH_Timing	8	6bh	SH_Timing.
69h	SH_Timing	8	6fh	SH_Timing.
6ah	SH_Timing	8	71h	SH_Timing.
6bh	SH_Timing	8	6dh	SH_Timing.
6ch	SH_Timing	8	a6h	SH_Timing.
6dh	SH_Timing	8	a7h	SH_Timing.
6eh	SH_Timing	7	09h	SH_Timing.
6fh	BRIGHT_ALL	7	00h	Bit[6]:The sign for brightness adjust: 1:positive 0:negative Bit[5:0]:value of brightness.
70h	IntCtr	8	0fh	Bit[7]:STOP_Sram--sram on/off 1'b0:on 1'b1:off Bit[6]:Raw_Switch--output rawdata select Bit[4]:G_off_En -- grid correction on/off 1'b0:off 1'b1:on Bit[3]:Edge_Switch--edge enhancement on/off 1'b0:off 1'b1:on Bit[2]:Clu_Switch--cluster correction on/off 1'b0:off 1'b1:on Bit[1]:Bp_Switch--bad pixel correction on/off 1'b0:off 1'b1:on

Address	Name	Width	Default value	Description
				Bit[0]:Lpf_Switch--denoise on/off 1'b0:off 1'b1:on
71h	<i>BpcCtr</i>	8	47h	Bit[7:4]:Bp_TH1--threshold for badpixel Bit[3]: Den_Out_En--denoise outdoor on/off 1'b0:on 1'b1:off Bit[2]: Denoise_sel--edge mean for denoise(on/off) 1'b0:select center pixel for denoise 1'b1:select edge mean for denoise Bit[1:0]:The gain value for cluster correction
72h	<i>DenCtr1</i>	8	27h	Bit[7:4]:Base_VAR Bit[3:0]:Base_gl_gain
73h	<i>DenCtr2</i>	8	68h	Bit[7:4]:Threshold to judge Bit[3:0]:Int_tim_t
74h	<i>DenCtr3</i>	8	a2h	Bit[7:4]:The flat region threshold value(*2) Bit[3]: Cen_Mean_sel Bit[1:0]:Edg_En_Max
75h	<i>DenCtr4</i>	8	89h	Bit[7:4]:Y_Th--Y_AVER threshold Bit[3:2]:Int_tim_cho Bit[0]: De_Low_En
76h	<i>DenCtr5</i>	8	8ah	Bit[4]: Den_Y_Aver_En Bit[3:0]: Den_Y_Aver_Th
77h	<i>EgeCtr1</i>	8	2fh	Bit[7:4]: Edge_TH--threshold for edge detection(*2) Bit[3:0]: Flat_TH--threshold for flat field(*2)
78h	<i>EgeCtr2</i>	8	23h	Bit[7:4]:Edge_Gain_P Bit[3:0]:Edge_Gain_N
79h	<i>EgeCtr3</i>	8	80h	Bit[7:5]: Skin denoise strength. Bit[4]: Gray_judge Bit[3]: Edg_Off_Step Bit[2]: Edg_Off_Mode Bit[1:0]:Sobel_Min
7ah	<i>EgeCtr4</i>	8	82h	Bit[7]: Sobel_en--thin edge function on/off 1'b0:off 1'b1:on Bit[4]: Low_Sh_Sw Bit[3:0]:Edg_Off_base
7bh	<i>SobMax</i>	8	ddh	{Bit[7],Bit[3]}:Gain_Denoise Bit[6:4]:Threshold for sobel difference($2^{\text{Bit}[6:4]}$) Bit[2:0]:Threshold for sobel difference($2^{\text{Bit}[2:0]}$)
7ch	<i>COM_SEL1</i>	8	97h	Bit[7:4]:Y_DE_TH Bit[3:0]:G_OFF_TH
7dh	<i>TARGET_SIGN1</i>	6	36h	Bit[5:0]: A light Color Correction Coefficients. Bit[5]: TARGET_SEL6. Bit[4]: TARGET_SEL5. Bit[3]: TARGET_SEL4. Bit[2]: TARGET_SEL3. Bit[1]: TARGET_SEL2. Bit[0]: TARGET_SEL1.



Address	Name	Width	Default value	Description
7eh	<i>TARGET_SIGN2</i>	6	16h	Bit[5:0]: Normal light Color Correction Coefficients. Bit[5]: TARGET_SEL6. Bit[4]: TARGET_SEL5. Bit[3]: TARGET_SEL4. Bit[2]: TARGET_SEL3. Bit[1]: TARGET_SEL2. Bit[0]: TARGET_SEL1.
80h	<i>AE_SPEED</i>	8	00h	Bit[7:4]: The speed of adjusting from light to dark Bit[3:0]: The speed of adjusting from dark to light
81h	<i>GLB_MIN1D</i>	8	1bh	GLB_MIN1 8 Bits
82h	<i>GLB_MAX1D</i>	8	37h	GLB_MAX1 8 Bits
83h	<i>GLB_MIN2D</i>	8	39h	GLB_MIN2 8 Bits(test)
84h	<i>GLB_MAX2D</i>	8	5dh	GLB_MAX2 8 Bits
85h	<i>GLB_MIN3D</i>	8	39h	GLB_MIN3 8 Bits
86h	<i>GLB_MAX3D</i>	8	30h	GLB_MAX3 8 Bits
87h	<i>GLB_GAIN0</i>	7	10h	GLB_GAIN0 register.
88h	<i>Y_AVER</i>	8	RO	The Y_aver of the current frame
89h	<i>INT_MID_I2C</i>	8	45h	Bit[7:4]: The lock threshold Bit[3:0]: INT_MID
8ah	<i>INT_STEP_50</i>	8	66h	The low 8 Bits of the minimum integral time for every step to avoid flicker for 50HZ/60HZ light.
8bh	<i>INT_STEP</i>	8	03h	Steps of INT_TIM
8ch	<i>INT_TIM[12:8]</i>	5	RO	Real integration time MSB.
8dh	<i>INT_TIM[7:0]</i>	8	RO	Real integration time LSB.
8fh	<i>INT_MIN</i>	8	82h	INT_MIN select.
90h	<i>GAIN_BEG_4D</i>	8	d7h	Curve line :the fifth beginning point
91h	<i>GAIN_END_4D</i>	8	30h	Curve line :the fifth ending point
92h	<i>GLB_HIGH</i>	7	88h	Bit[6:4]:The threshold of gain. Bit[3]: The ending point of line 4,the msb. Bit[2]: The starting point of line 4,the msb. Bit[1]: The ending point of line 3,the msb. Bit[0]: The starting point of line 3,the msb.
94h	<i>TAR_BASE0</i>	8	62h	Bit[7:4]:(192+Bit[7:4]*4) as threshold to judge one pixel whether to be over exposure pixel. Bit[3:0]:Control the start of AE.
95h	<i>TAR_BASE1</i>	8	88h	Bit[7:4]: is used to modify the difference of Y_AVER and modified AE_TAR) .the smaller TAR_BASE1[7:4] is ,the slower the AE adjusting. Bit[3:0]:The smallest value the target brightness can achieve.(AE_TAR*TAR_BASE1[3:0]/16).the smaller TAR_BASE1[3:0] is,the quicker the AE adjusting.
99h	<i>DIG_GAIN_I2C</i>	8	10h	The value of DIG_GAIN.
9ah	<i>INT_TIM_TH</i>	8	18h	Threshold for INT_TIME to judge outdoor scene.
9dh	<i>Gain_OR_Last</i>	8	RO	The value of gain_or_last[8:1] to get the GLB_GAIN0 through some operation



Address	Name	Width	Default value	Description
9eh	<i>YBRIGHT_TH</i>	8	b6h	Bit[7:4]: AE adjusting. Bit[3:0]: Limit the speed of AE
9fh	<i>AE_MODE DIG_GAIN_MAX</i>	8	14h	Bit[7]: Reserved. Bit[6]: Control the adjustive speed of digital gain 1'b0 :Y_DIFF_D[6:1] (1 time) 1'b1 :Y_DIFF_D (2 times) Bit[5]: Setps enable. Bit[4]: AE is adjusted by men, Bit[3:0]: DIG_GAIN_MAX
a0h	<i>AWB_CTR_SET</i>	8	09h	Bit[7]: UPDATE_MODE Bit[6]: J_CTRL6,sign white pixel selection Bit[5]: J_CTRL5, Bit[4]: J_CTRL4,combination1 selection. Bit[3]: J_CTRL3 Bit[2]: J_CTRL2,PIXEL_WP_EN setting. Bit[1]: J_CTRL1,combination2 selection. Bit[0]: AWB_ENABLE: 1'b1:AWB on 1'b0:AWB off
a1h	<i>AWB_TH1_SET</i>	8	10h	Bit[7:4]:AWB_LOCK, Auto white balance lock boundary. Bit[3:2]:SKIN_BGAIN_TH_L. Bit[1:0]:SKIN_RGAIN_TH_L.
a2h	<i>BLU_GAIN_TH1</i>	6	0bh	Bit[7:6]: Reserved. Bit[5:0]: Minimum blue gain.
a3h	<i>BLU_GAIN_TH2</i>	6	30h	Bit[7:6]: Reserved. Bit[5:0]: Maximum blue gain.
a4h	<i>RED_GAIN_TH1</i>	6	09h	Bit[7:6]: Reserved. Bit[5:0]: Minimum red gain.
a5h	<i>RED_GAIN_TH2</i>	6	30h	Bit[7:6]: Reserved. Bit[5:0]: Maximum red gain.
a6h	<i>COUNT_EN</i>	8	56h	Bit[7:4]: Region 3 counter threshold. Bit[3:0]: White pixel counter threshold.
a7h	<i>BASE_B_GAIN</i>	6	12h	Bit[7:6]:Reserved. Bit[5:0]:Base B gain.
a8h	<i>BASE_R_GAIN</i>	6	12h	Bit[7:6]:Reserved. Bit[5:0]:Base R gain.
a9h	<i>AWB_GB_LIM</i>	8	12h	Bit[7:6]: Reserved. Bit[5:0]:AWB criterion: B
aah	<i>AWB_GR_LIM</i>	8	12h	Bit[7:6]: Reserved. Bit[5:0]:AWB criterion: R
abh	<i>AWB_BR_LIM</i>	8	18h	AWB criterion: BR,for region 1.
ach	<i>AWB_Y_TH</i>	8	a8h	AWB criterion
adh	<i>AWB_G_LIM</i>	8	12h	The threshold for region 3.
aeh	<i>F_GAIN_TH_H</i>	8	59h	Bit[7:4]: B limit to estimate F light. Bit[3:0]: R limit to estimate F light.



Address	Name	Width	Default value	Description
afh	<i>DIG_SLOPE</i>	8	0ah	Bit[7:4]: Color Gain offset Bit[3:0]: Slop of color gain for digital gain
b0h	<i>SAT_CTRL1</i>	8	00h	Bit[7:4]: Select the mean of the gray section Bit[5:0]: Low light ctrl.
b1h	<i>CB_COEF_NF</i>	8	c6h	Cb Saturation Coefficient low 8 bit for NF
b2h	<i>CR_COEF_NF</i>	8	cch	Cr Saturation Coefficient low 8 bit for NF
b3h	<i>CB_COEF_F</i>	8	d6h	Cb Saturation Coefficient low 8 bit for F
b4h	<i>CR_COEF_F</i>	8	dch	Cr Saturation Coefficient low 8 bit for F
b5h	<i>SAT_GAIN_TH</i>	5	0ch	Bit[4:0]:The threshold of GLB_GAIN.
b6h	<i>TEST_MODE</i>	8	00h	Bit[7:0]: 8'h00: Normal output. Bit[7:0]: 8'h01~8'h0f : Fixed vertical bar pattern. Bit[7:0]: 8'h10~8'h1f : Fixed horizontal bar pattern. Bit[7:0]: 8'h20~8'h2f : Fixed vertical gradual pattern. Bit[7:0]: 8'h30~8'h3f : Fixed horizontal gradual pattern. Bit[7:0]: 8'h40~8'h5f : Fixed manual pattern. Bit[7:0]: 8'h60~8'h7f: Normal output.
b7h	<i>MAN_R</i>	8	80h	Bit[7:0]: Define R/G/B value.
b8h	<i>GAIN_DIFF COLOR_CTRL</i>	8	93h	Bit[7:4]: Gain_diff (Used to wipe off gain function). Bit[1]: Pure blue enable Bit[0]: Pure red enable
b9h	<i>SKIN_GAIN_TH</i>	8	64h	Skin select.
bah	<i>AWB_CB_LIM1</i>	8	12h	Bit[7:6]: Reserved. Bit[5:0]:AWB criterion: B.
bah	<i>AWB_CR_LIM1</i>	8	12h	Bit[7:6]: Reserved. Bit[5:0]: AWB criterion: R.
bch	<i>SH_Timing</i>	8	08h	SH_Timing.
bdh	<i>SH_Timing</i>	8	04h	SH_Timing.
beh	<i>SH_Timing</i>	8	07h	SH_Timing.
bfh	<i>SH_Timing</i>	8	09h	SH_Timing.
c0h	<i>SH_Timing</i>	8	0ch	SH_Timing.
c1h	<i>SH_Timing</i>	8	10h	SH_Timing.
c2h	<i>SH_Timing</i>	8	15h	SH_Timing.
c3h	<i>SH_Timing</i>	8	11h	SH_Timing.
c4h	<i>SH_Timing</i>	8	14h	SH_Timing.
c5h	<i>SH_Timing</i>	8	16h	SH_Timing.
c6h	<i>SH_Timing</i>	8	19h	SH_Timing.
c7h	<i>SH_Timing</i>	8	01h	SH_Timing.
c8h	<i>SH_Timing</i>	6	2ah	SH_Timing.
c9h	<i>SH_Timing</i>	6	2ch	SH_Timing.
cah	<i>SH_Timing</i>	8	1ah	SH_Timing.
cbh	<i>LINE_LENGTH_OUT[7:0]</i>	8	90h	Bit[7:0] :LINE_LENGTH[7:0].
cch	<i>{5'B0,LINE_LENGTH_OUT[10:8]}</i>	8	01h	Bit[7:3]: Reserved. Bit[2:0] :LINE_LENGTH[12:8].
cdh	<i>SH_Timing</i>	8	16h	SH_Timing.



Address	Name	Width	Default value	Description
ceh	<i>SH_Timing</i>	8	14h	SH_Timing.
cfh	<i>SH_Timing</i>	8	17h	SH_Timing.
d0h	<i>SH_Timing</i>	8	30h	SH_Timing.
d1h	<i>SH_Timing</i>	8	32h	SH_Timing.
d2h	<i>S SH_Timing</i>	8	7bh	SH_Timing.
d5h	<i>CON_CTRL1</i>	6	08h	Bit[5]: Gray section denoise switch. 1'b0:disable 1'b1:enable Bit[4]: Gray section denoise mean select. Bit[3:0]: Bright, increase brightness in low light scene.
d6h	<i>Y_COEF</i>	8	80h	Y Coefficient for Contrast
d8h	<i>GAIN_BEG_0D</i>	8	0fh	Curve line :the first beginning point
d9h	<i>GAIN_END_0D</i>	8	20h	Curve line :the first ending point
dah	<i>GAIN_BEG_1D</i>	8	21h	Curve line :the second beginning point
dbh	<i>GAIN_END_1D</i>	8	42h	Curve line :the second ending point
dch	<i>GAIN_BEG_2D</i>	8	43h	Curve line :the third beginning point
ddh	<i>GAIN_END_2D</i>	8	7eh	Curve line :the third ending point
deh	<i>GAIN_BEG_3D</i>	8	84h	Curve line :the forth beginning point
dfh	<i>GAIN_END_3D</i>	8	d6h	Curve line :the forth ending point
e0h	<i>REG_PD</i>	8	01h	Bit[7:1]: Reserved. Bit[0]: Soft PD(Analog ctrl). 1'b0: Power on. 1'b1: Power down.
e1h	<i>REG_DIV</i>	8	92h	Bit[7:5]: CLK SEL. Bit[4]: VCLK invert. 1'b0:not invert. 1'b1: invert(default). Bit[3:0]: CLK SEL
e2h	<i>REG_DIV2</i>	8	01h	Bit[7:5]: Reserved. Bit[4:3]: VCLK delay. Bit[2:0]: Reserved.
e3h	<i>REG_PLL</i>	8	82h	Bit[7]: PLL Bypass. 1'b0: Do not bypss. Fout=Fpll/ z1 1'b1: Bypass pll(default) . Fout=Fin. Bit[6]: OEN_VSYNC_CLK. 1'b0:Enable. 1'b1:Disable. Bit[5:4]: Post Div for PLL(Fout) 2'b00: Div 1(z=1)(default). 2'b01: Div 2(z=2). 2'b10: Div 4(z=4). 2'b11: Div 8(z=8). Bit[3:2]: Reserved. Bit[1:0]: PLL Pre_sel[1:0]. 2'b00: x=2. 2'b01: x=4. 2'b10: x=8(default). 2'b11: x=9. Fref=Fin/x=3MHz.
e4h	<i>REG_BIAS</i>	8	92h	Reserved.



Address	Name	Width	Default value	Description
e5h	<i>REG_BIAS2</i>	8	22h	Reserved.
e6h	<i>REG_COM</i>	8	24h	Reserved.
e7h	<i>REG_COM2</i>	8	66h	Bit[7]: OEN_HSYNC_DATA. 1'b0:Enable. 1'b1:Disable. Bit[6:0]: Reserved.
e8h	<i>REG_COM3</i>	8	a1h	Bit[7]: Reserved. Bit[6:5]: Sel for DATA drive. Bit[4]: Sel for I2C SDA drive. Bit[3:0]: LDO Sel.
eeh	<i>P_TH</i>	6	0fh	Bit[5:0]: Probability Threshold
efh	<i>SKIN_CTR</i>	3	4h	Bit[2]: 1'b0: Disable skin function, 1'b1: Enable skin function. Bit[1]: 1'b0: Disable white pixels signing function, 1'b1: Enable white pixels signing function. Bit[0]: 1'b0: display full resolution, 1'b1: only display skin area.
f0h	<i>INT_MAX_I2C</i>	6	53h	Bit[7:6]: P_OE_SEL Bit[5:0]:INT_MAX, the MAX steps of integral time
f1h	<i>ISPBYP</i>	7	00h	Bit[7]:Reserved Bit[6]:Contrast enable 1'b0: enable, 1'b1: disable Bit[5]:Saturation enable 1'b0: enable, 1'b1: disable Bit[4]:CC_SP enable 1'b0: enable, 1'b1: disable Bit[3]:Color Correction enable 1'b0: enable, 1'b1: disable Bit[2]:Color Intrpolation enable 1'b0: enable, 1'b1: disable Bit[1]:GammaCorrection enable 1'b0: enable, 1'b1: disable Bit[0]:LensCorrection enable 1'b0: enable, 1'b1: disable
f2h	<i>SCCB_RESET</i>	1	0h	Bit[0]:SCCB_RESET 1'b0: No change 1'b1: Resets all registers to default values
f3h	<i>BURRFILT</i>	8	01h	Parameter for I2C start signal filter
fbh	<i>VER_BME</i>	4	RO	Bit[3:0] : Product ver..
fch	<i>PIDH_BME</i>	8	20h,RO	Bit[7:0] : Product ID MSB.
fdh	<i>PIDL_BME</i>	8	a6h,RO	Bit[7:0] : Product ID LSB.

8. Package Specifications

8.1 CSP Package

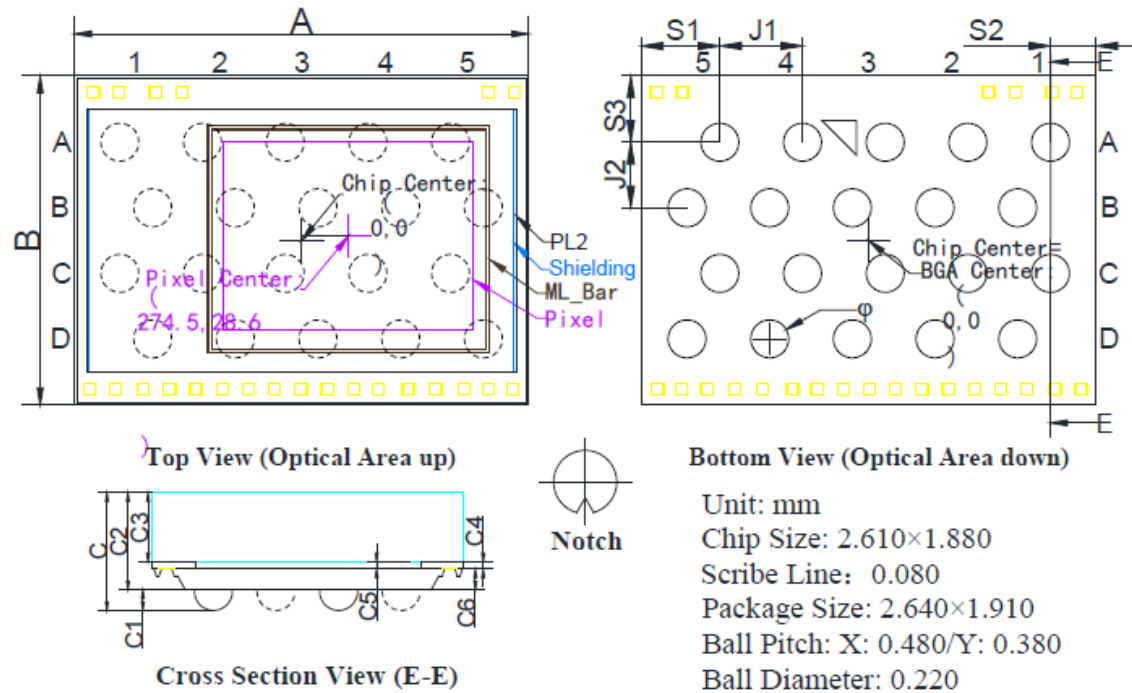


Figure 7 CSP dimension description

Table 7. CSP Dimensions

页面导航	Symbol	Nominal	Min	Max	Nominal	Min	Max
meter		Millimeters			Inches		
Package Body Dimension X	A	2.6400	2.6150	2.6650	0.103937	0.102953	0.104921
Package Body Dimension Y	B	1.9100	1.8850	1.9350	0.075197	0.074213	0.076181
Package Height	C	0.6900	0.6350	0.7450	0.027165	0.025000	0.029331
Ball Height	C1	0.1200	0.0900	0.1500	0.004724	0.003543	0.005906
Package Body Thickness	C2	0.5700	0.5400	0.6000	0.022441	0.021260	0.023622
Glass Thickness	C3	0.4000	0.3900	0.4100	0.015748	0.015354	0.016142
Cavity Wall Height	C4	0.0300	0.0260	0.0340	0.001181	0.001024	0.001339
Cavity Wall+ Epoxy Thickness	C5	0.0325	0.0275	0.0375	0.001280	0.001083	0.001476
Si Thickness	C6	0.1000	0.0950	0.1050	0.003937	0.003740	0.004134
Ball Diameter	Φ	0.2200	0.1900	0.2500	0.008661	0.007480	0.009843
Total Ball Count	N	20	/	/	20	/	/
Ball Count X Axis	N1	5	/	/	5	/	/
Ball Count Y Axis	N2	4	/	/	4	/	/
Ball Pitch X Axis 1	J1	0.4800	0.4700	0.4900	0.018898	0.018504	0.019291
Ball Pitch Y Axis 2	J2	0.3800	0.3700	0.3900	0.014961	0.014567	0.015354
Package Center to Chip Center Offset in X-Direction(Optical Area up)	ΔX	0.0000	-0.0250	0.0250	0.000000	-0.000984	0.000984
Package Center to Chip Center Offset in X-Direction(BGA Area up)	ΔY	0.0000	-0.0250	0.0250	0.000000	-0.000984	0.000984
BGA Ball Center to Chip Center Offset in X-Direction(BGA Area up)	ΔX1	0.0000	-0.0250	0.0250	0.000000	-0.000984	0.000984
BGA Ball Center to Chip Center Offset in Y-Direction(BGA Area up)	ΔY1	0.0000	-0.0250	0.0250	0.000000	-0.000984	0.000984
Package Body Edge to Ball Center Distance along X1	S1	0.4552	0.4252	0.4852	0.017921	0.016740	0.019102
Package Body Edge to Ball Center Distance along X2	S2	0.2648	0.2348	0.2948	0.010425	0.009244	0.011606
Package Body Edge to Ball Center Distance along Y	S3	0.3850	0.3550	0.4150	0.015157	0.013976	0.016339

Table 8. CSP Pin Descriptions

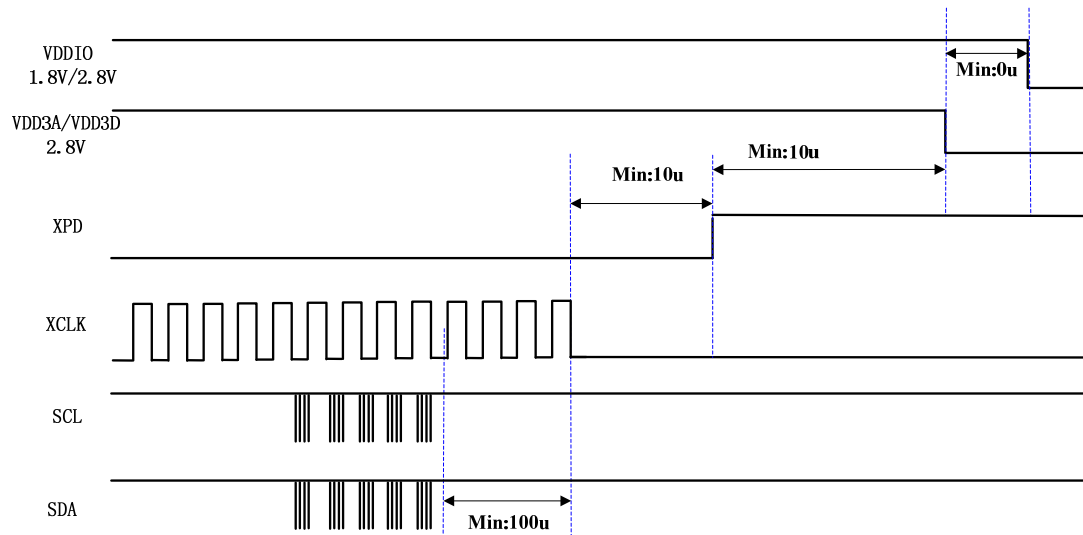
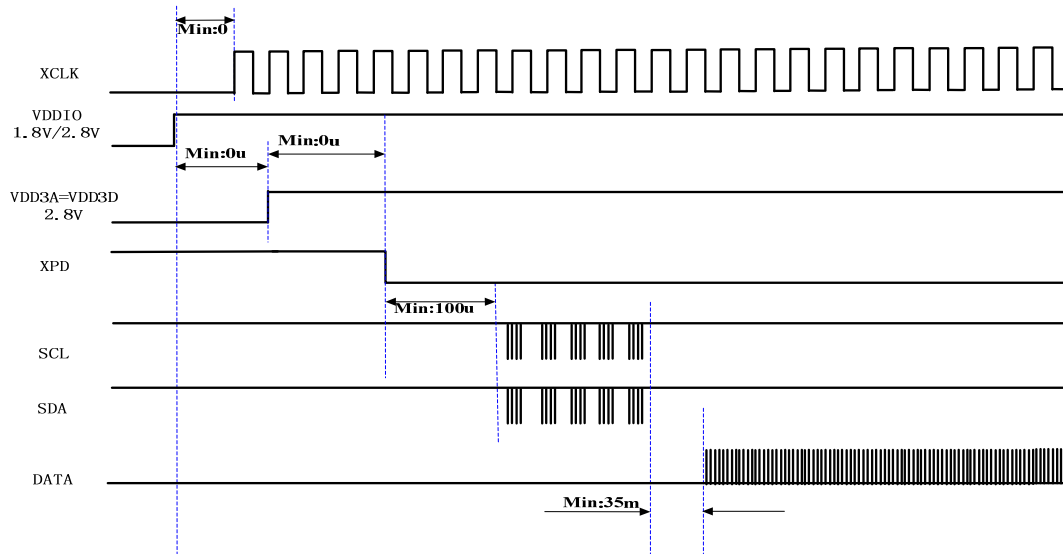


Pin Number	Name	Pin Type	Function/Description
A1	VDDIO	Supply	I/O power supply:1.7~ 3.1V
A2	VCLK	Output	DVP or SPI clock output
A3	VSYNC	Output	Vertical synchronization output
A4	SCL	Input	SCCB serial interface clock input.
A5	VDD3A	Supply	Analog power 2.8V
B1	HSYNC	Output	Horizontal synchronization output
B2	D[2]	Output	DVP D[2] or SPI D[2] output
B3	D[5]	Output	DVP D[5] output
B4	XCLK	Input	System clock input.
B5	VSSA	Ground	Analog ground
C1	VSSD	Ground	Digital ground.
C2	D[1]	Output	DVP D[1] or SPI D[1] output
C3	D[4]	Output	DVP D[4] output
C4	D[7]	Output	DVP D[7] output
C5	XPD	Input	Power Down Mode Selection. 0: Normal mode. 1: Power down mode.
D1	D[0]	Output	DVP D[0] or SPI D[0] output
D2	D[3]	Output	DVP D[3] or SPI D[3] output
D3	D[6]	Output	DVP D[6] output
D4	SDA	I/O	SCCB serial interface data I/O .
D5	VDD3D	Supply	Digital power 2.8V

Note:

1. XPD : pull-down to VSSD by inner resistor.

9. Application Timing Diagram





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